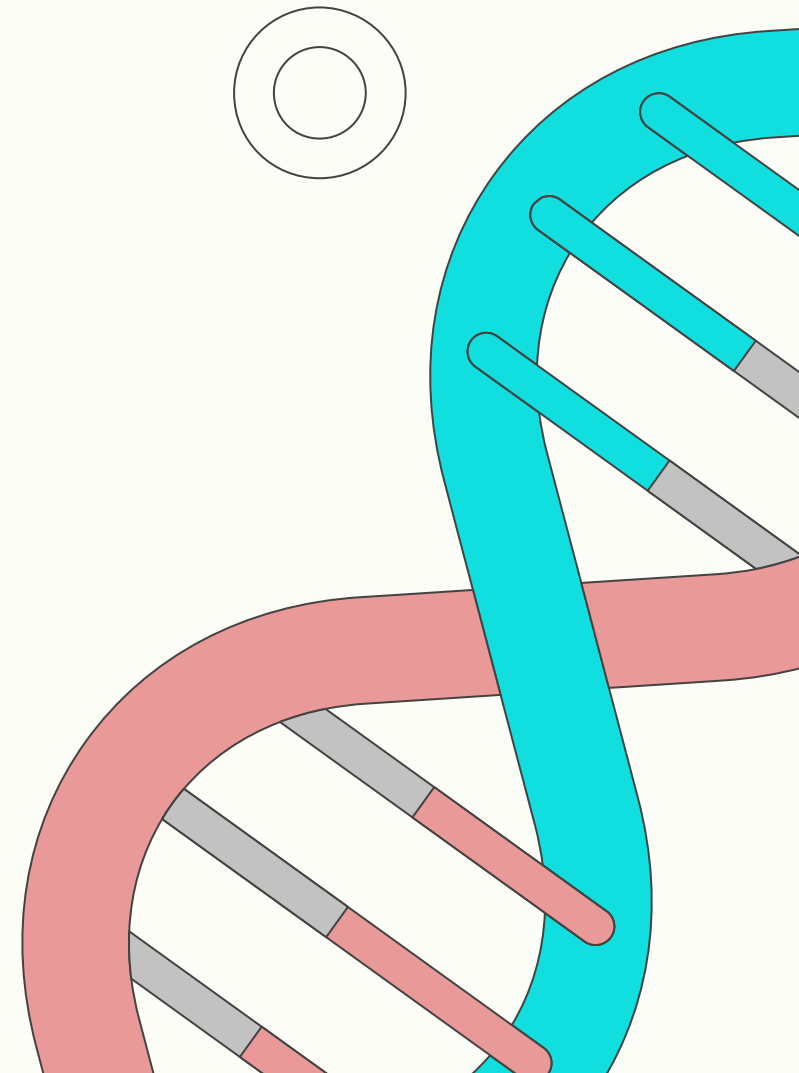
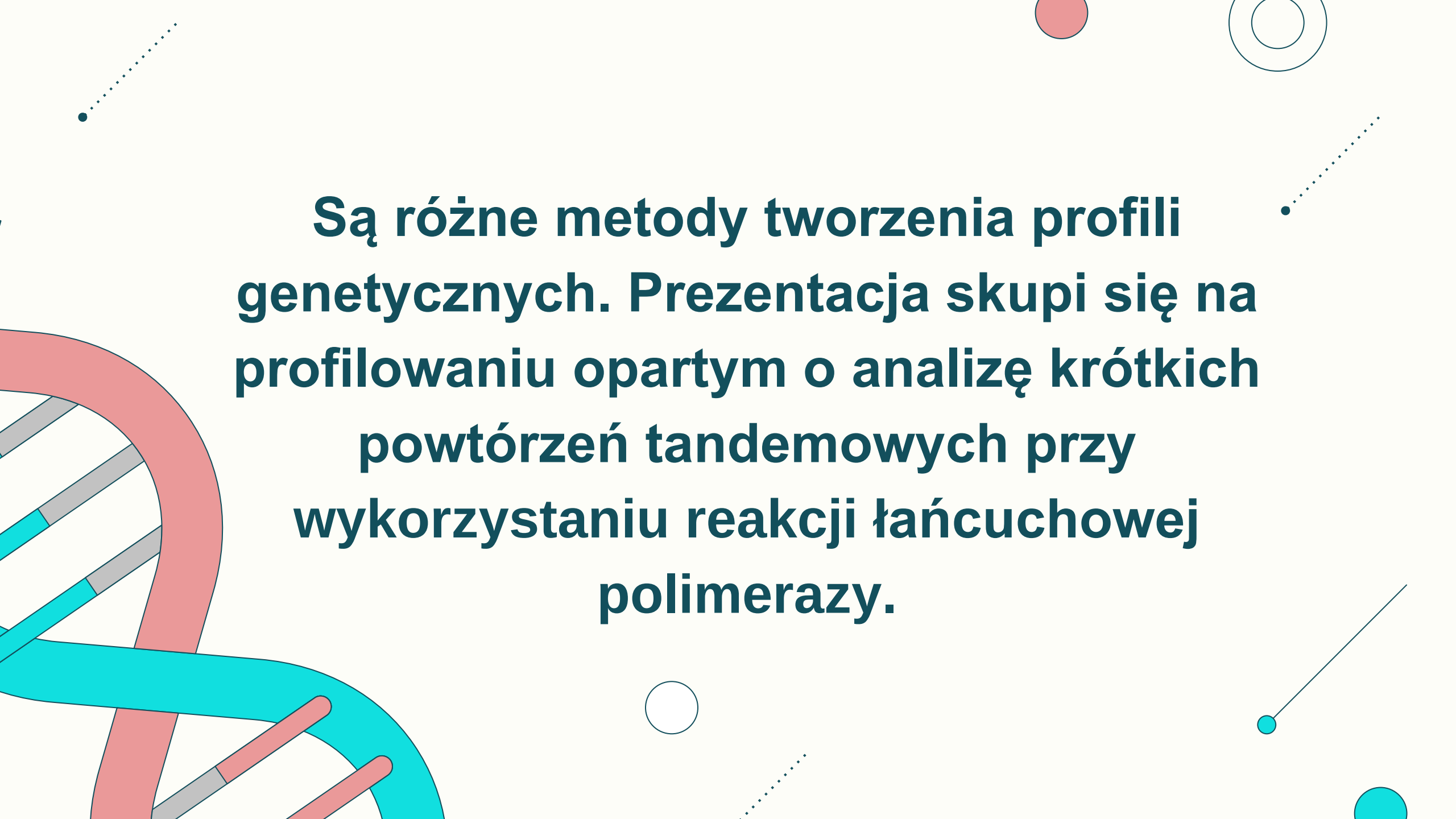


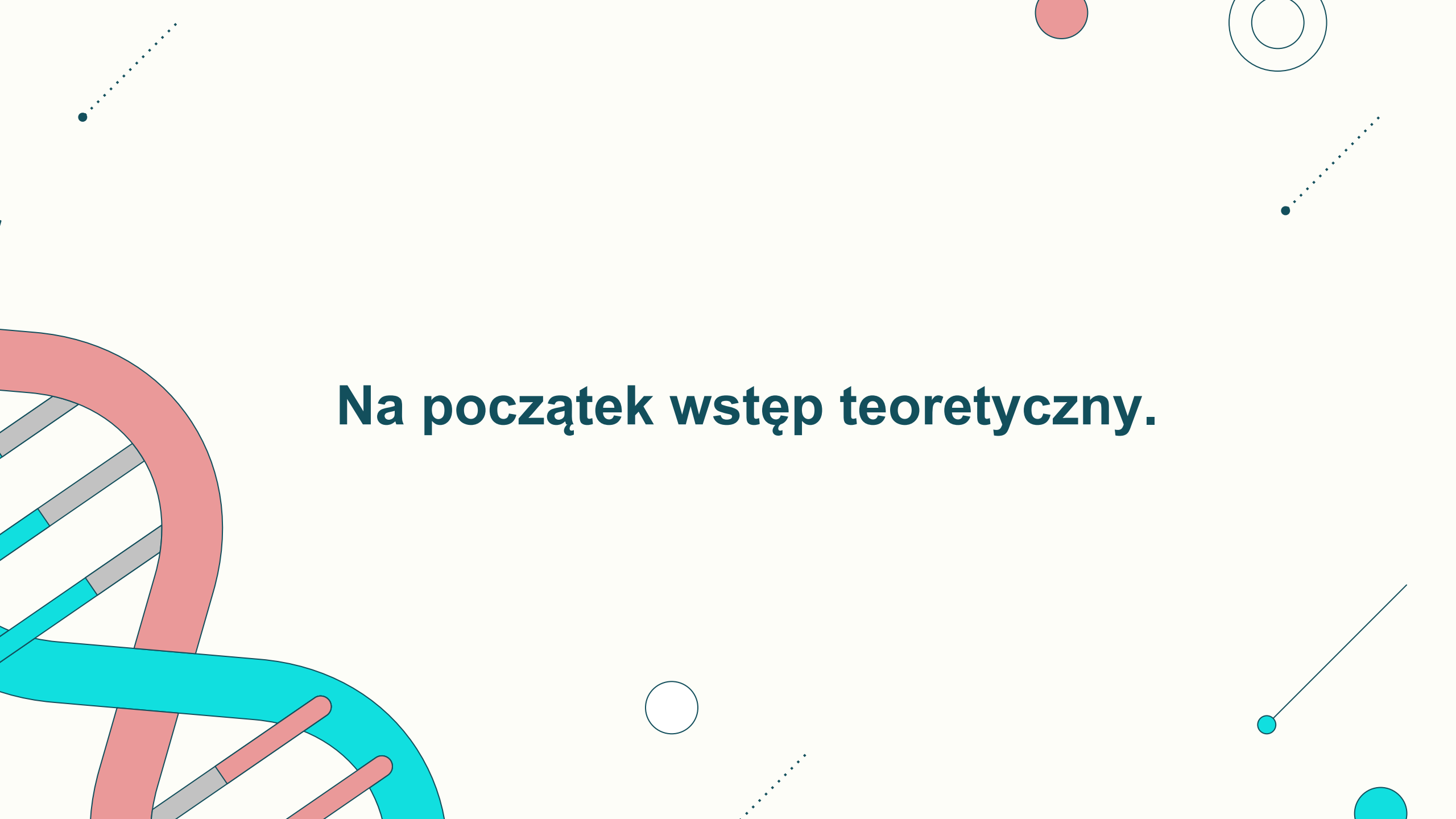
Biometria DNA

Sebastian Pergała



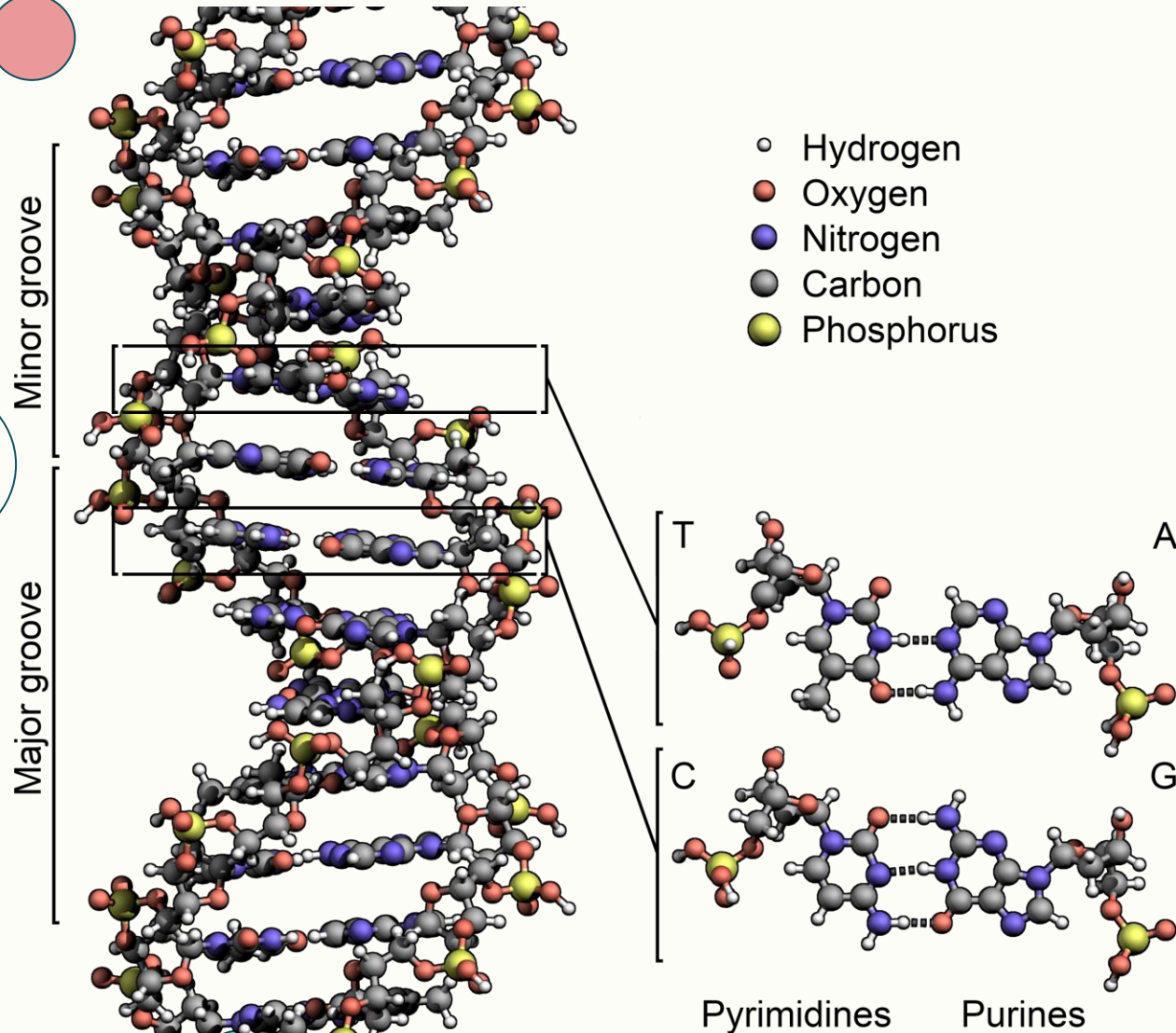


Są różne metody tworzenia profili genetycznych. Prezentacja skupi się na profilowaniu opartym o analizę krótkich powtórzeń tandemowych przy wykorzystaniu reakcji łańcuchowej polimerazy.

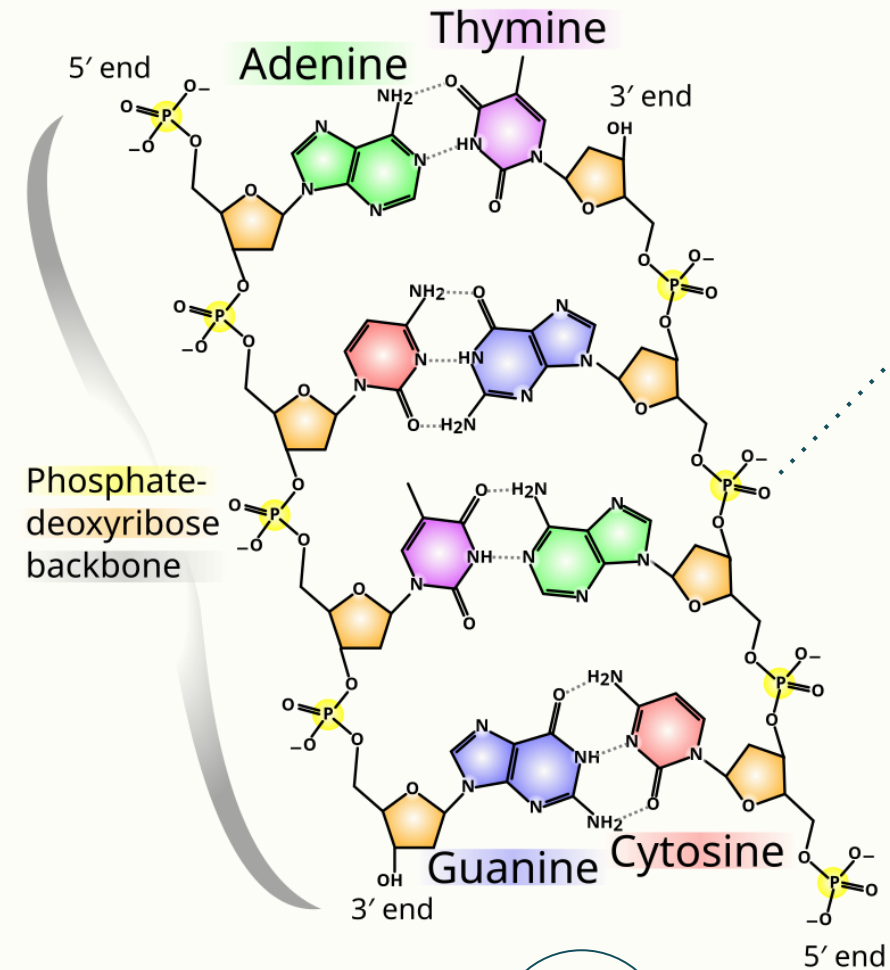
The background is a light gray with several abstract geometric elements. In the top left, there is a dotted line with a solid black dot at its end. In the top right, there is a solid red circle and a set of concentric circles (an outer blue circle and an inner white circle). In the bottom left, a large, stylized DNA double helix is partially visible, with red and cyan bands and gray rungs. In the bottom right, there is a solid cyan circle, a dotted line with a solid cyan dot, and a thin blue line with a solid cyan dot.

Na początek wstęp teoretyczny.

Czym jest DNA?

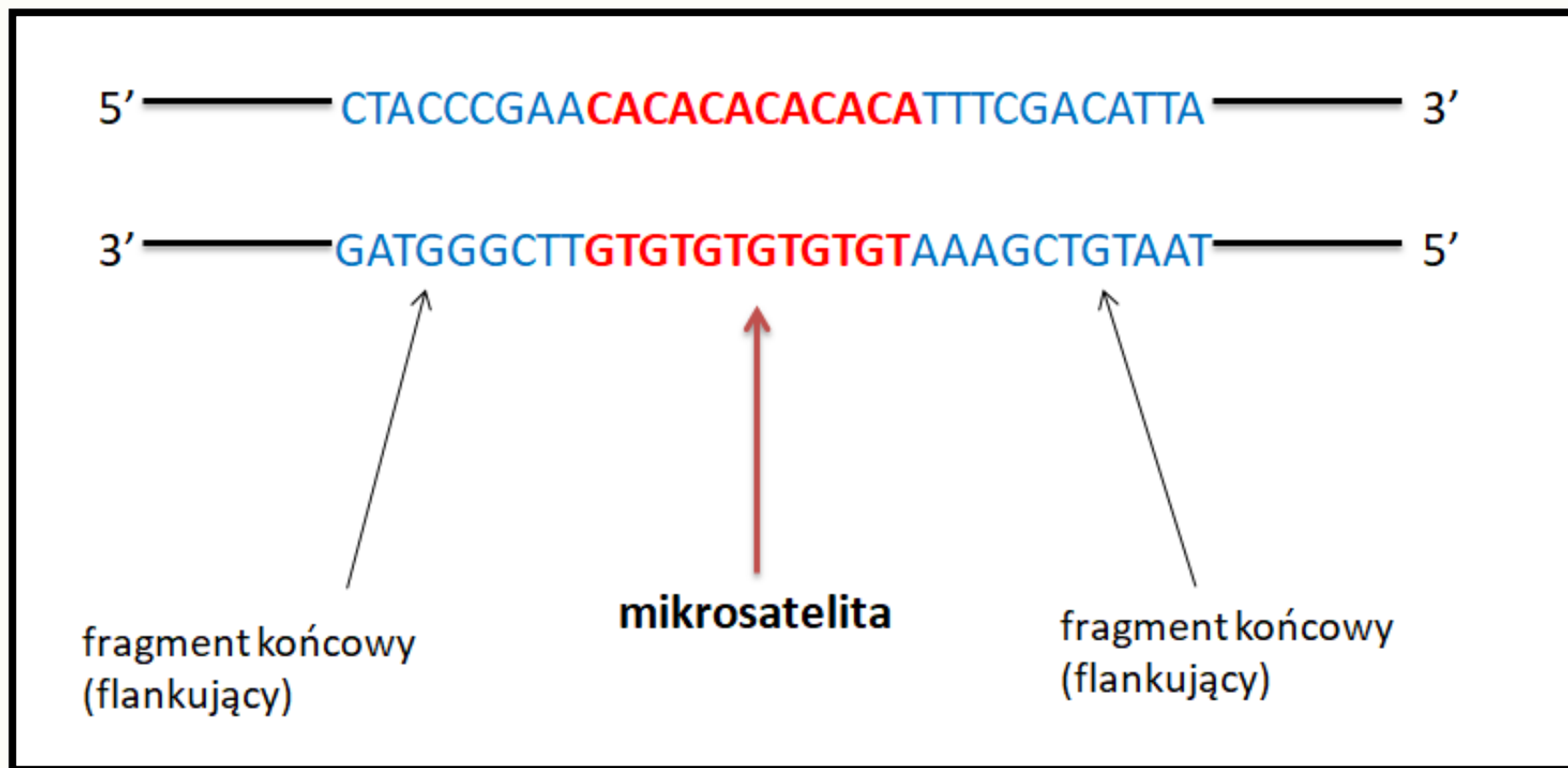


https://en.wikipedia.org/wiki/DNA#/media/File:DNA_Structure+Key+Labelled.pn_NoBB.png



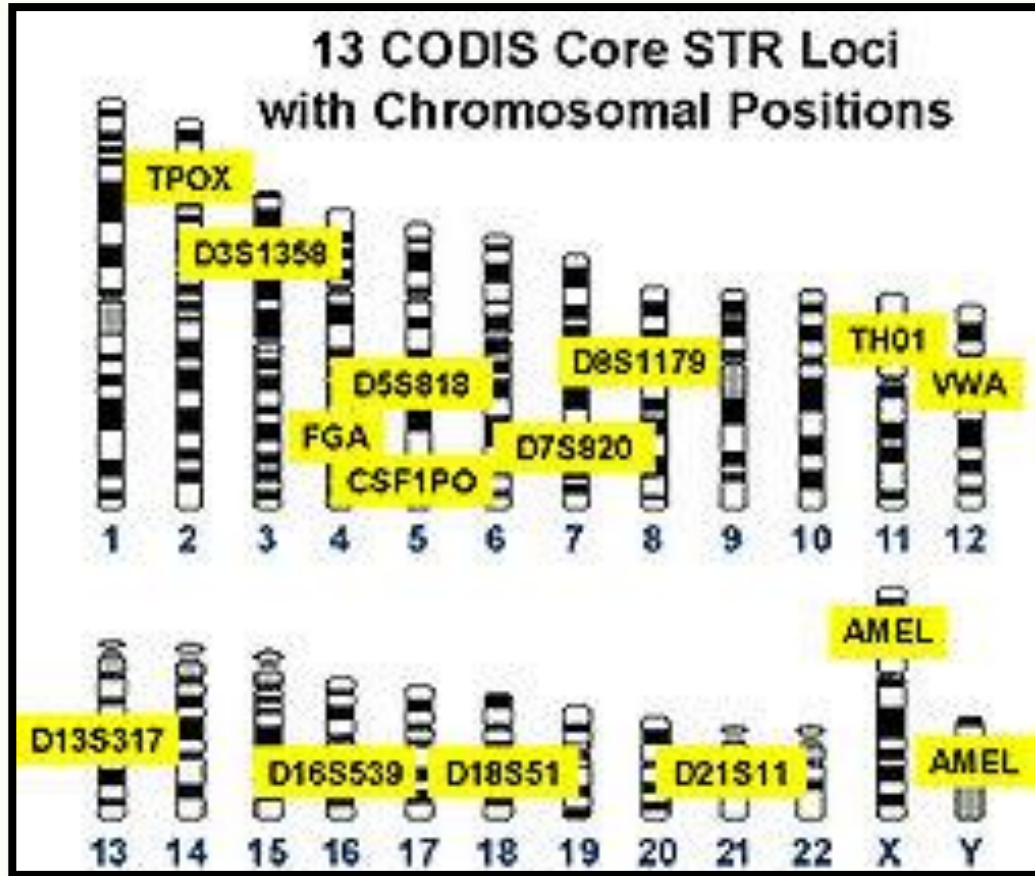
https://en.wikipedia.org/wiki/DNA#/media/File:DNA_chemical_structure.svg

Sekwencje mikrosatelitarne

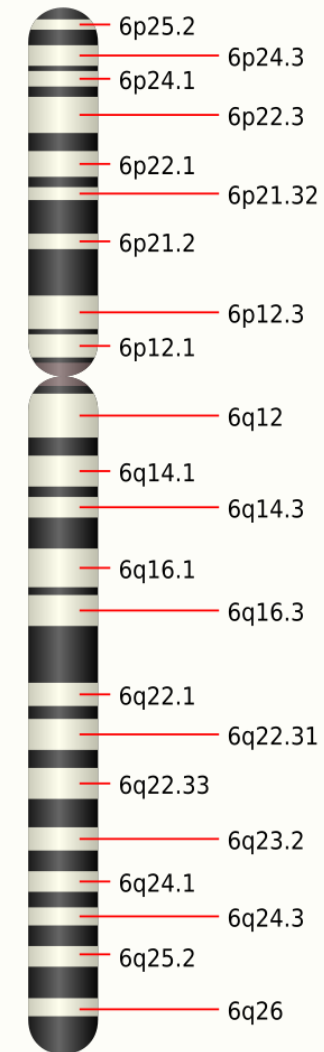


https://pl.wikipedia.org/wiki/Sekwencje_mikrosatelitarne#/media/Plik:Budowa_loci_mikrosatelitarnych.png

Markery

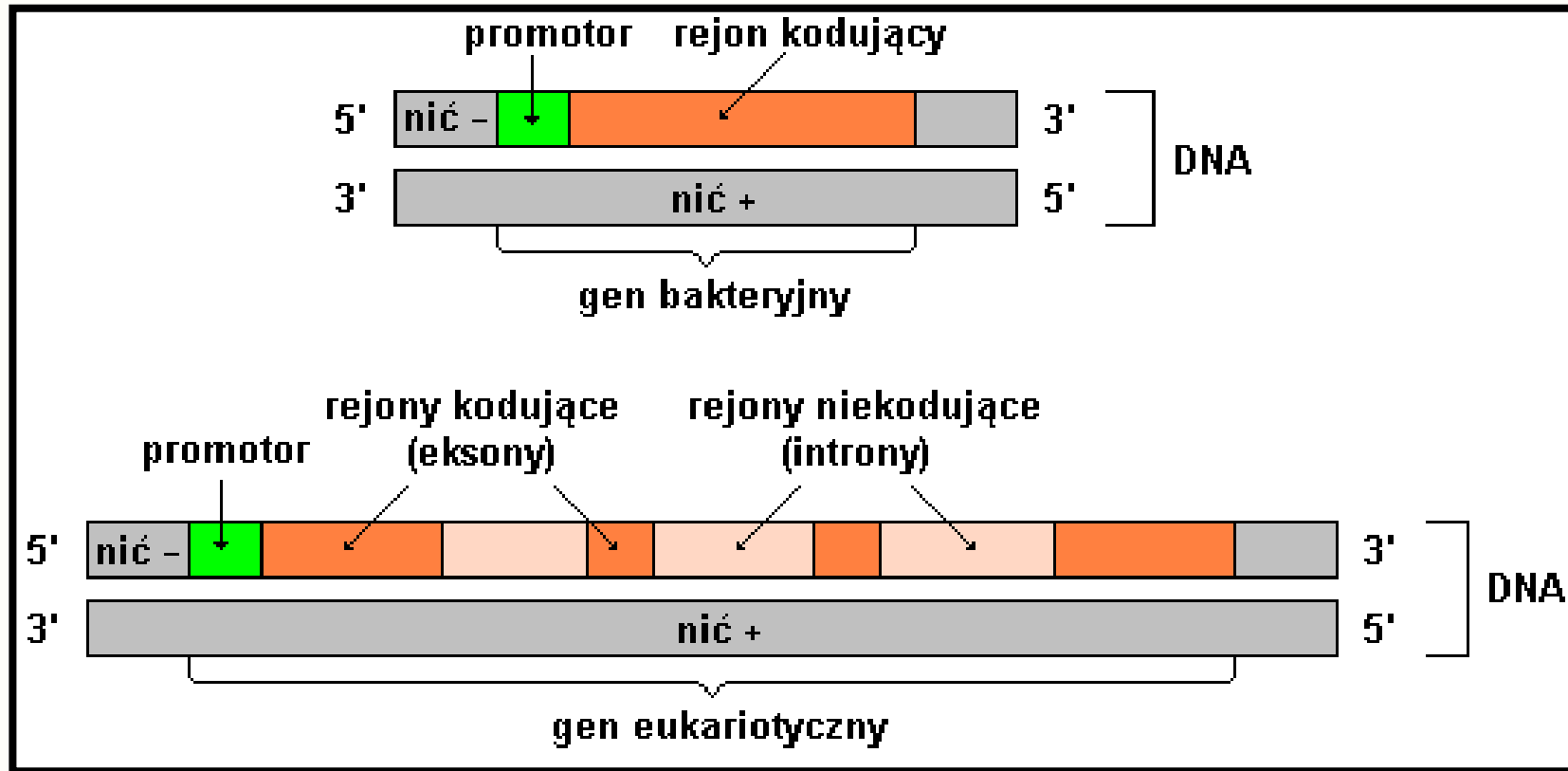


https://en.wikipedia.org/wiki/Combined_DNA_Index_System#/media/File:Codis_profile.jpg

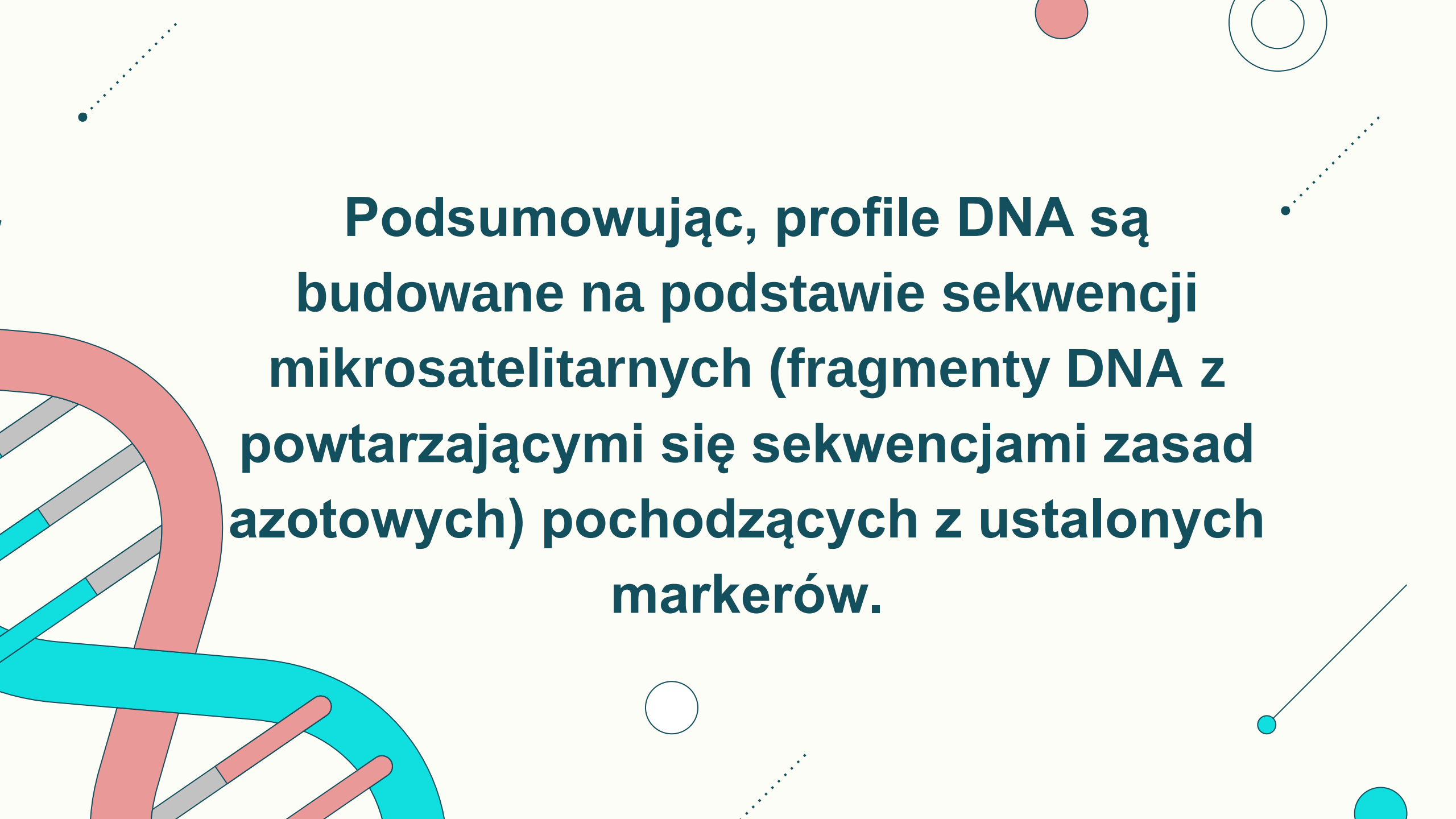


https://pl.wikipedia.org/wiki/Locus#/media/Plik:Chromosome_6.svg

Regiony kodujące i niekodujące



https://pl.wikipedia.org/wiki/Intron#/media/Plik:Por%C3%B3wnanie_gen%C3%B3w.PNG



**Podsumowując, profile DNA są
budowane na podstawie sekwencji
mikrosatelitarnych (fragmenty DNA z
powtarzającymi się sekwencjami zasad
azotowych) pochodzących z ustalonych
markerów.**

The background is a light gray with various geometric elements. In the top left, a dotted line with a black dot extends from the edge. In the top right, there is a solid red circle, a set of concentric circles (an outer blue line and an inner white circle), and another dotted line with a black dot. On the left side, a portion of a DNA double helix is visible, with red and cyan bands and gray rungs. In the bottom right, there is a solid cyan circle, a dotted line with a black dot, and a thin blue line with a cyan dot.

**Jak otrzymać z pełnego DNA
interesujące nas fragmenty DNA?**

Reakcja łańcuchowa polimerazy

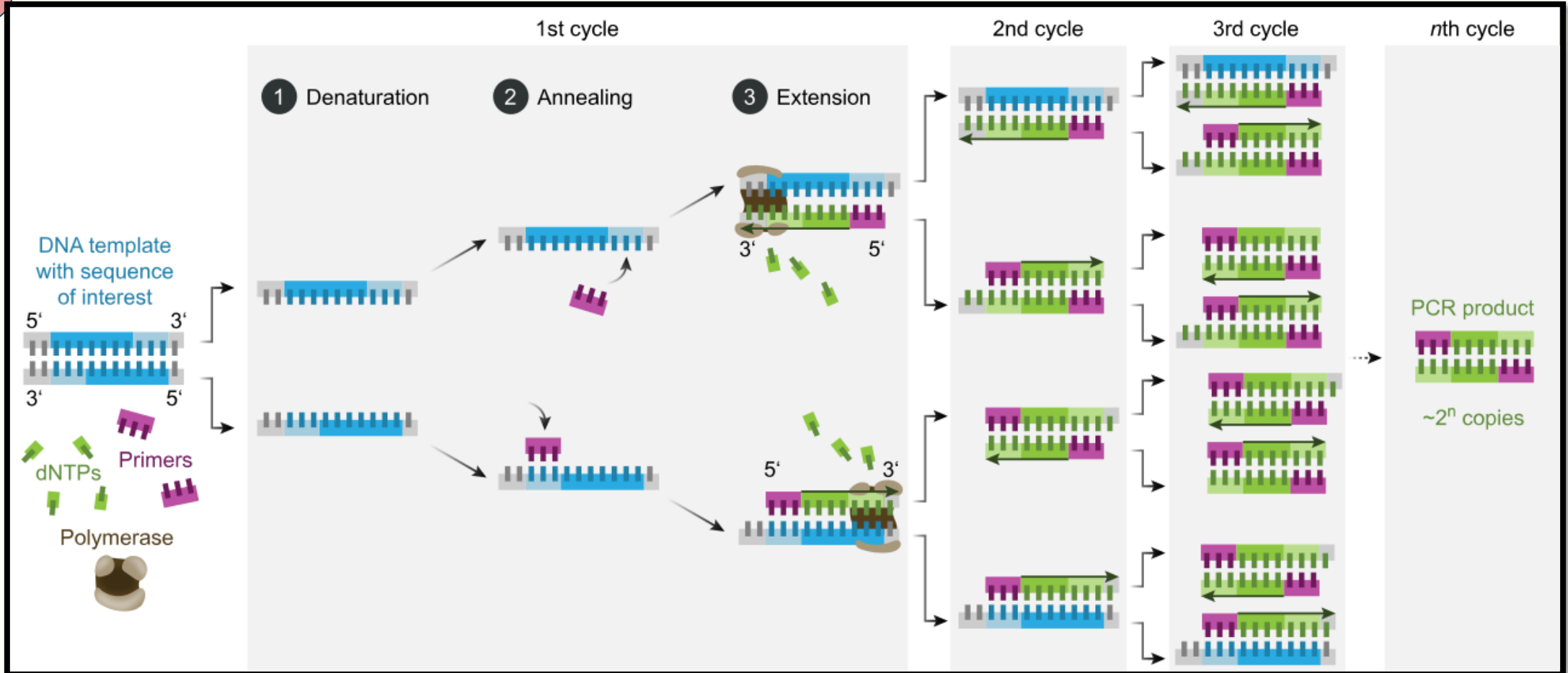


https://en.wikipedia.org/wiki/Polymerase_chain_reaction#/media/File:PCR_tubes.png

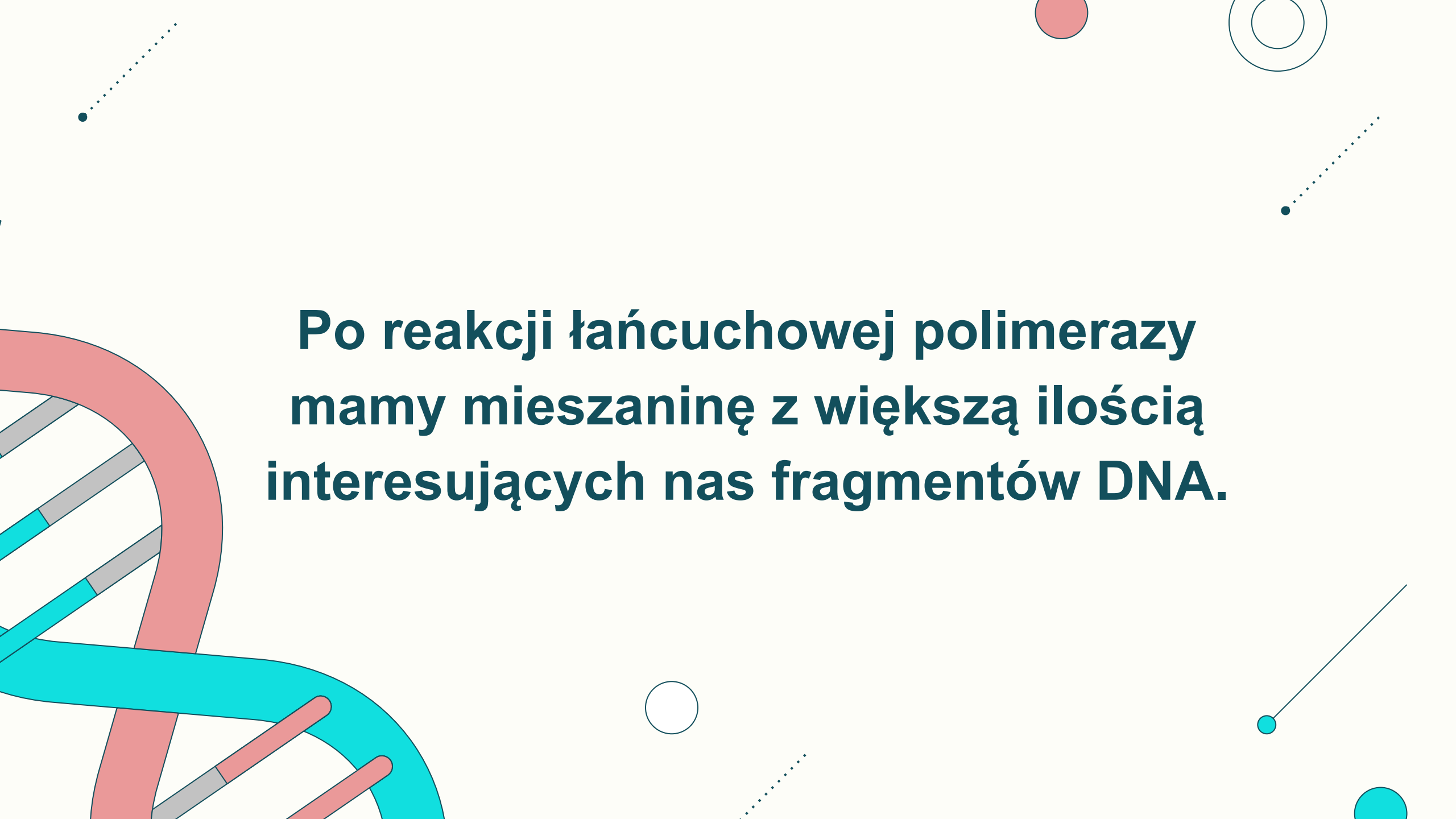


https://pl.wikipedia.org/wiki/Reakcja_%C5%84cuchowa_polimerazy#/media/Plik:Eppendorf_Mastercycler_Pro_S.jpg

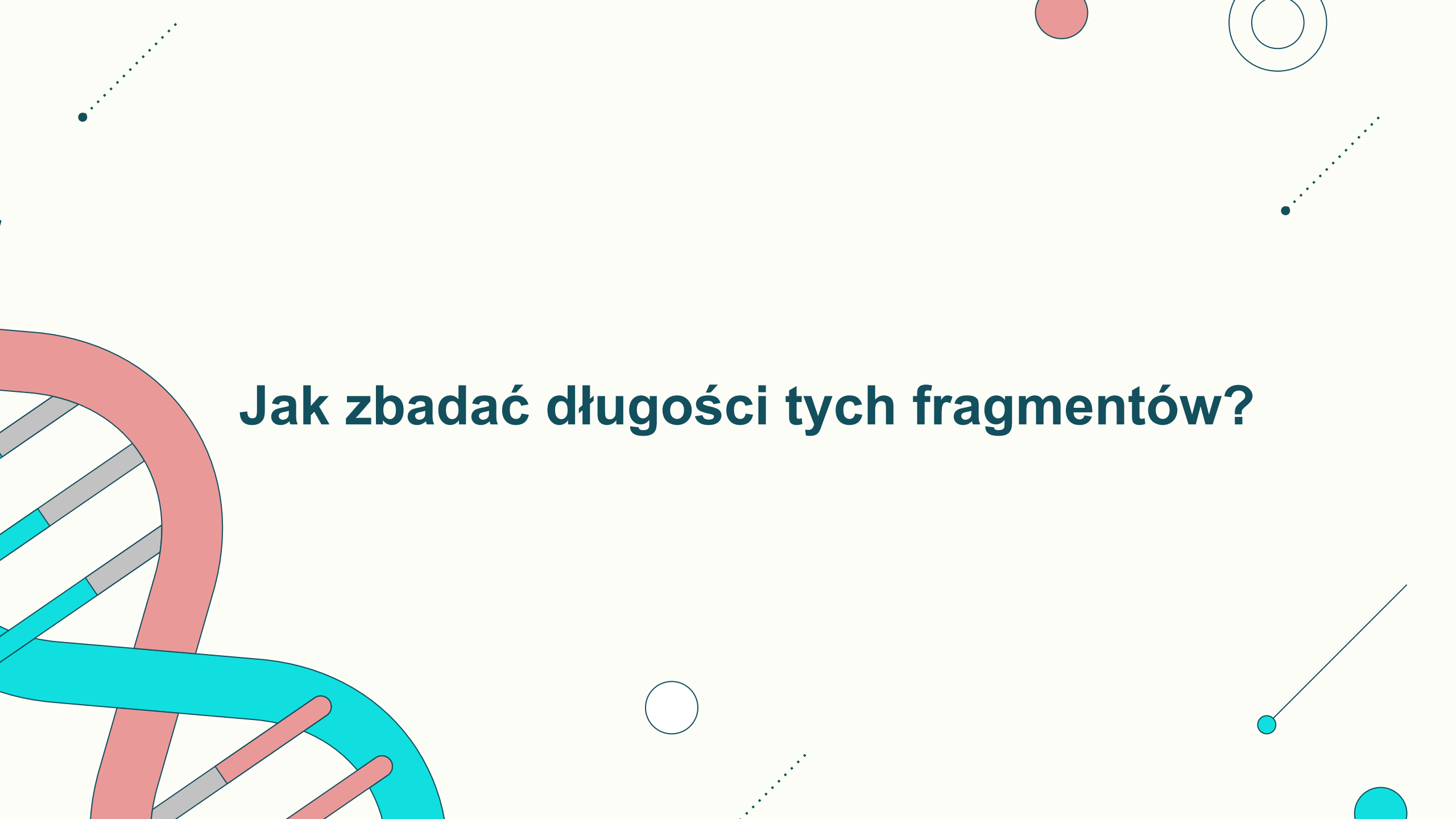
Reakcja łańcuchowa polimerazy



https://en.wikipedia.org/wiki/Polymerase_chain_reaction#/media/File:Polymerase_chain_reaction-en.svg

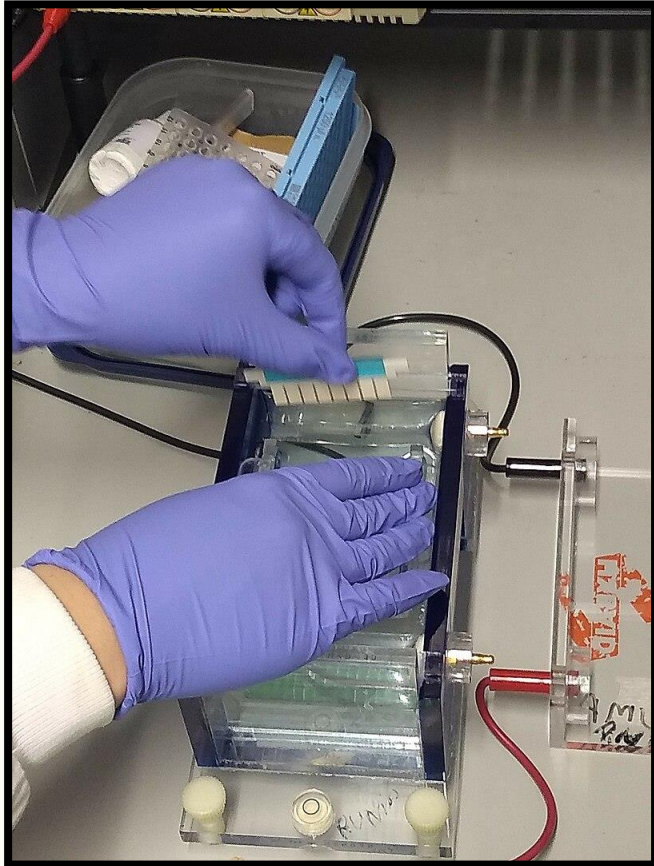
The background features a light gray color with various geometric elements. In the top left, there is a dotted line with a black dot. In the top right, there is a solid red circle and a thin-lined circle. In the bottom left, a portion of a DNA double helix is visible, with red and cyan strands and gray rungs. In the bottom right, there is a thin solid line with a cyan dot, a small cyan circle, and a dotted line with a black dot.

**Po reakcji łańcuchowej polimerazy
mamy mieszaninę z większą ilością
interesujących nas fragmentów DNA.**

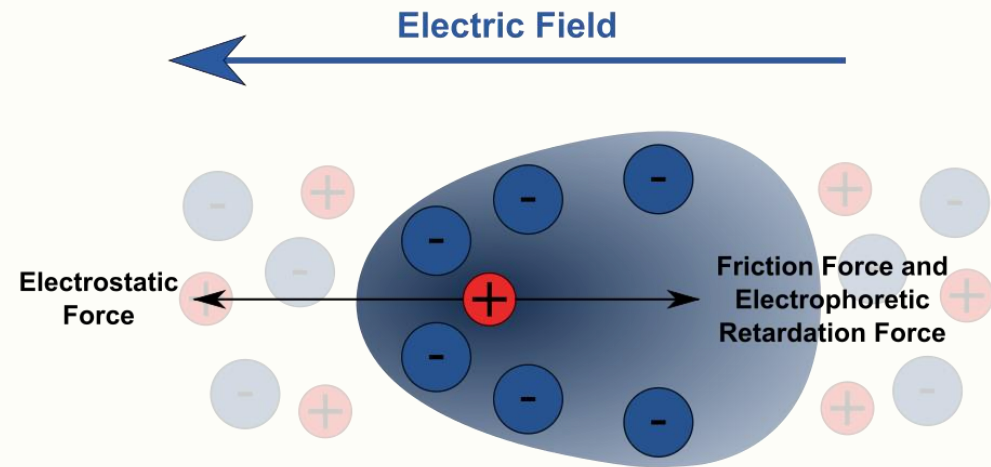
The background is a light gray with several abstract geometric elements. In the top left, there is a dotted line with a solid black dot at its end. In the top right, there is a solid red circle and a set of two concentric circles. In the bottom left, a large, stylized DNA double helix is partially visible, with red and cyan bands and gray rungs. In the bottom right, there is a solid cyan circle, a dotted line with a solid cyan dot, and a thin solid line with a solid cyan dot.

Jak zbadać długości tych fragmentów?

Elektroforeza

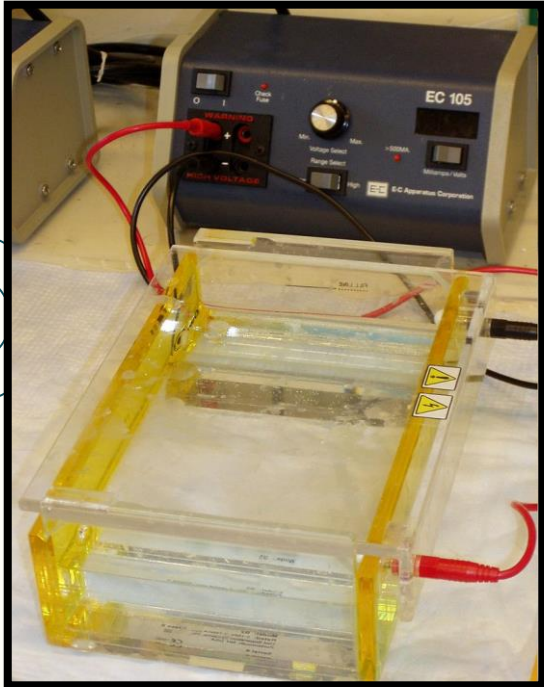


https://en.wikipedia.org/wiki/Gel_electrophoresis#/media/File:Gel_electrophoresis_insert_comb.jpg

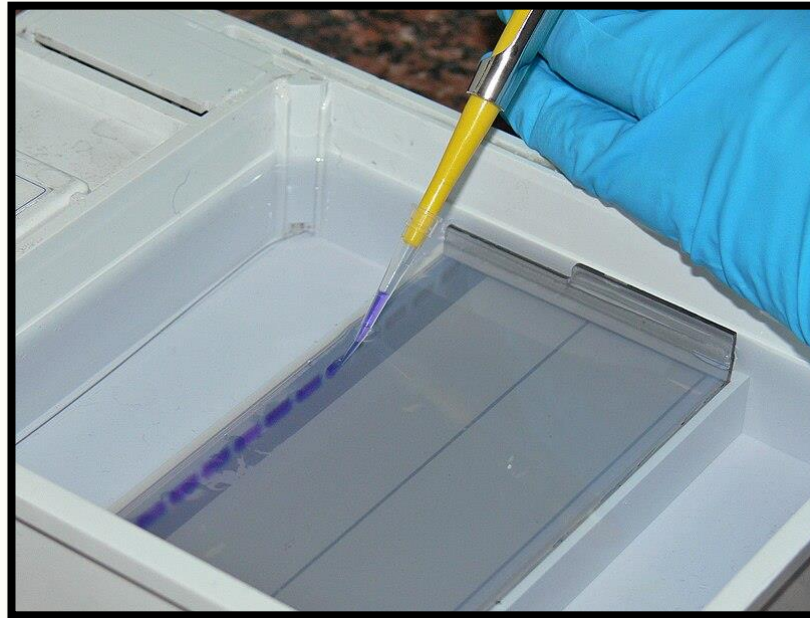


https://en.wikipedia.org/wiki/Electrophoresis#/media/File:Motion_by_electrophoresis_of_a_charged_particle.svg

Elektroforeza dla DNA



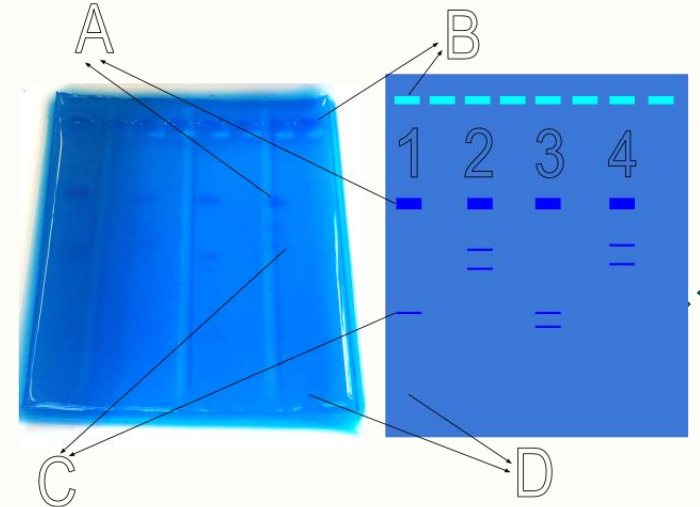
https://pl.wikipedia.org/wiki/Elektroforeza#/media/Plik:Gel_elektrophoresis_apparatus.JPG



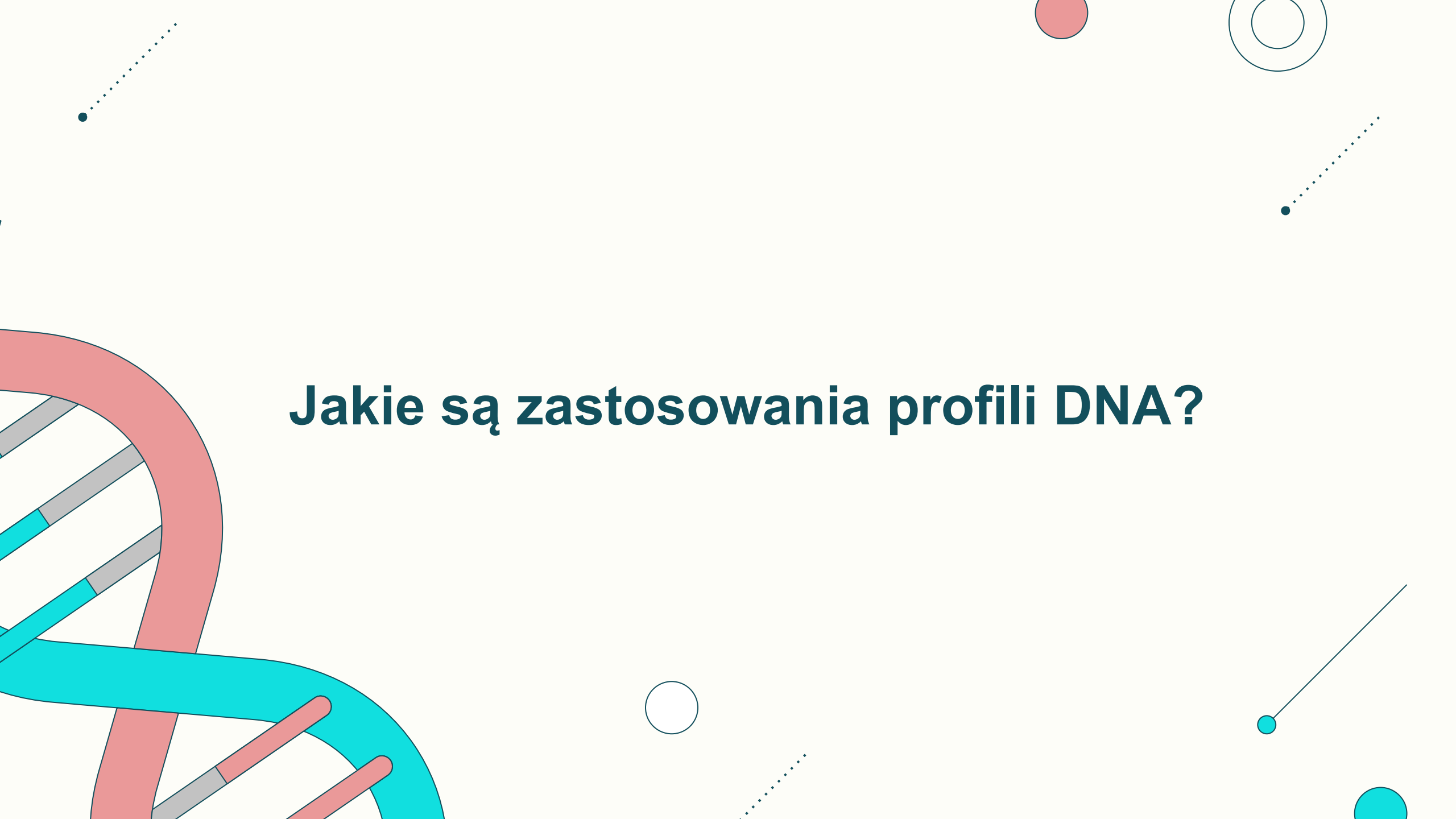
<https://pl.wikipedia.org/wiki/Elektroforeza#/media/Plik:Elektrof.jpg>

DNA Profiling

RFLP (Restriction Fragment Length Polymorphism) Analysis



https://pl.wikipedia.org/wiki/Profilowanie_DNA#/media/Plik:DNA_Profiling_-_RFLP_Analysis.svg

The background is a light gray with various geometric elements. In the top left, a dotted line with a black dot extends from the top left towards the center. In the top right, there is a solid red circle, a set of concentric circles (an outer blue line and an inner white circle), and another dotted line with a black dot extending from the top right towards the center. On the left side, a portion of a DNA double helix is visible, with red and cyan bands and gray rungs. In the bottom right, there is a solid cyan circle, a dotted line with a black dot extending from the bottom right towards the center, and a thin blue line with a cyan dot extending from the bottom right towards the center.

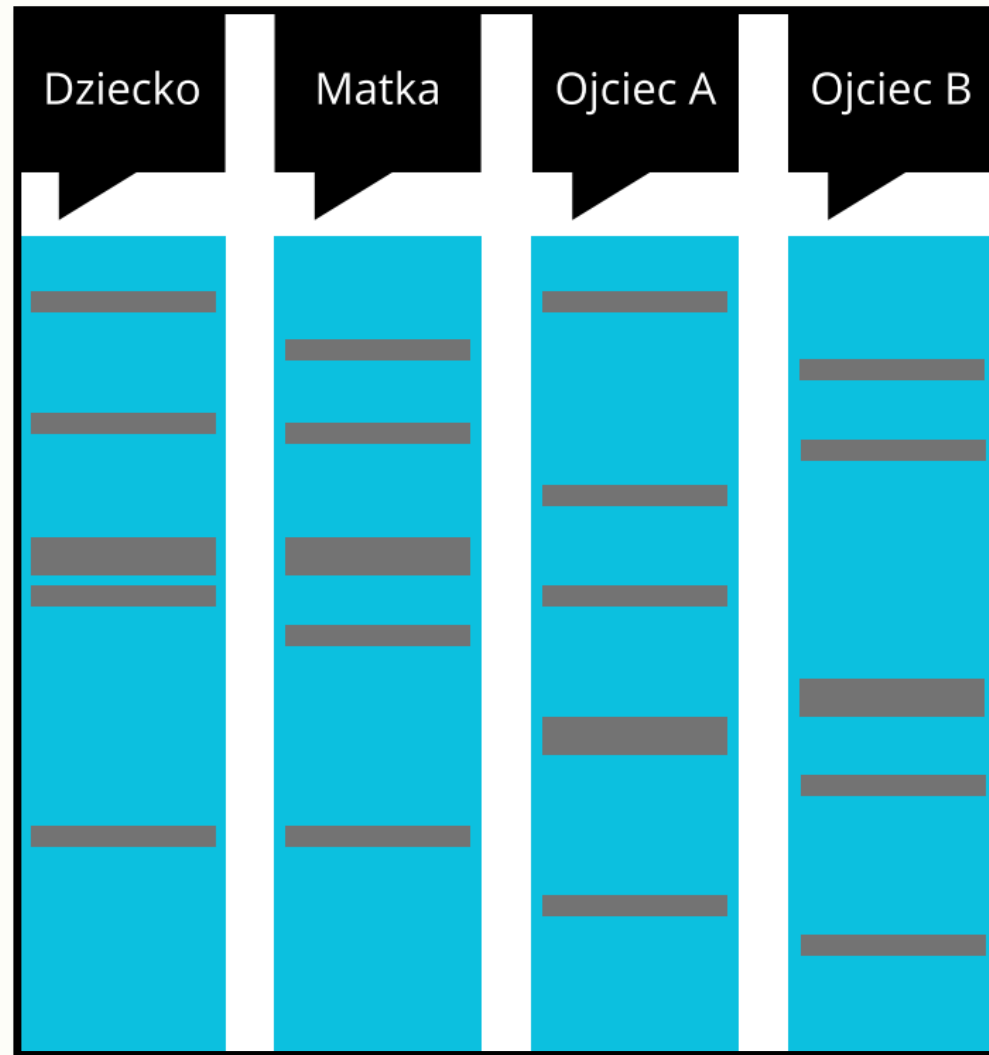
Jakie są zastosowania profili DNA?

[illegible]

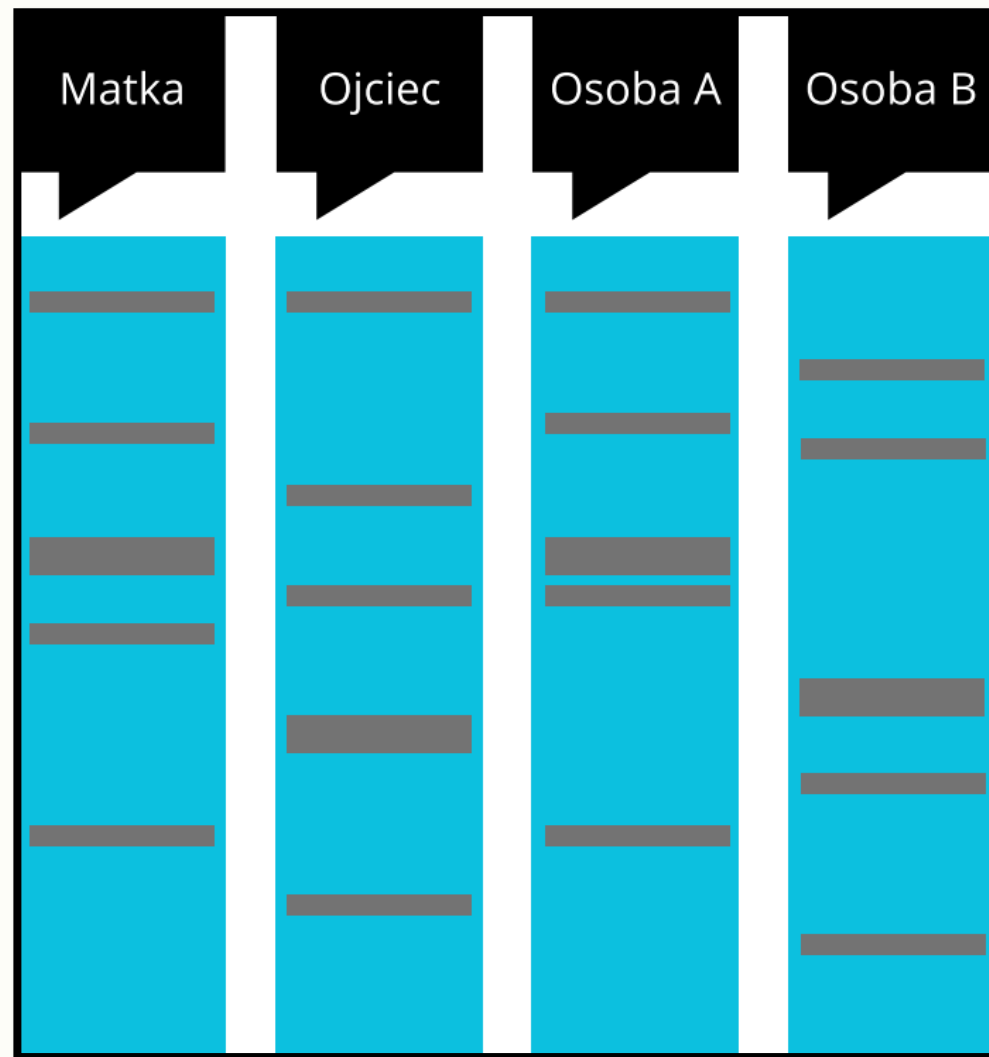
Sprawca	Ofiara	Podejrzany A	Podejrzany B
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
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99	99	99	99
100	100	100	100

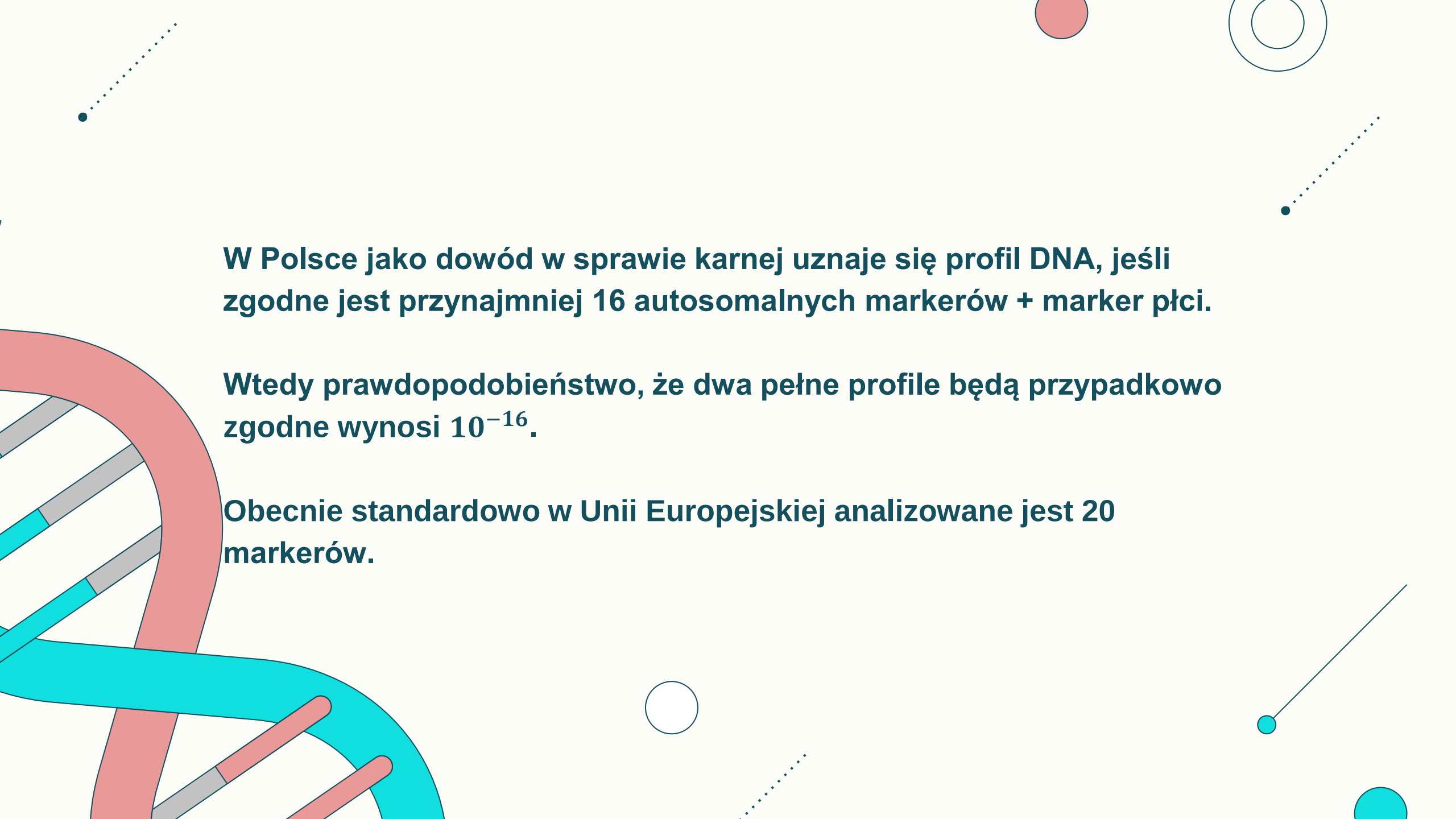
Ustalanie macierzyństwa / ojcostwa

Dziecko	Matka	Ojciec A	Ojciec B
Blue	Blue	Blue	Blue
Green	Green	Green	Green
Red	Red	Red	Red
Orange	Orange	Orange	Orange
Purple	Purple	Purple	Purple
Pink	Pink	Pink	Pink
Grey	Grey	Grey	Grey



Ustalanie tożsamości osoby zaginionej

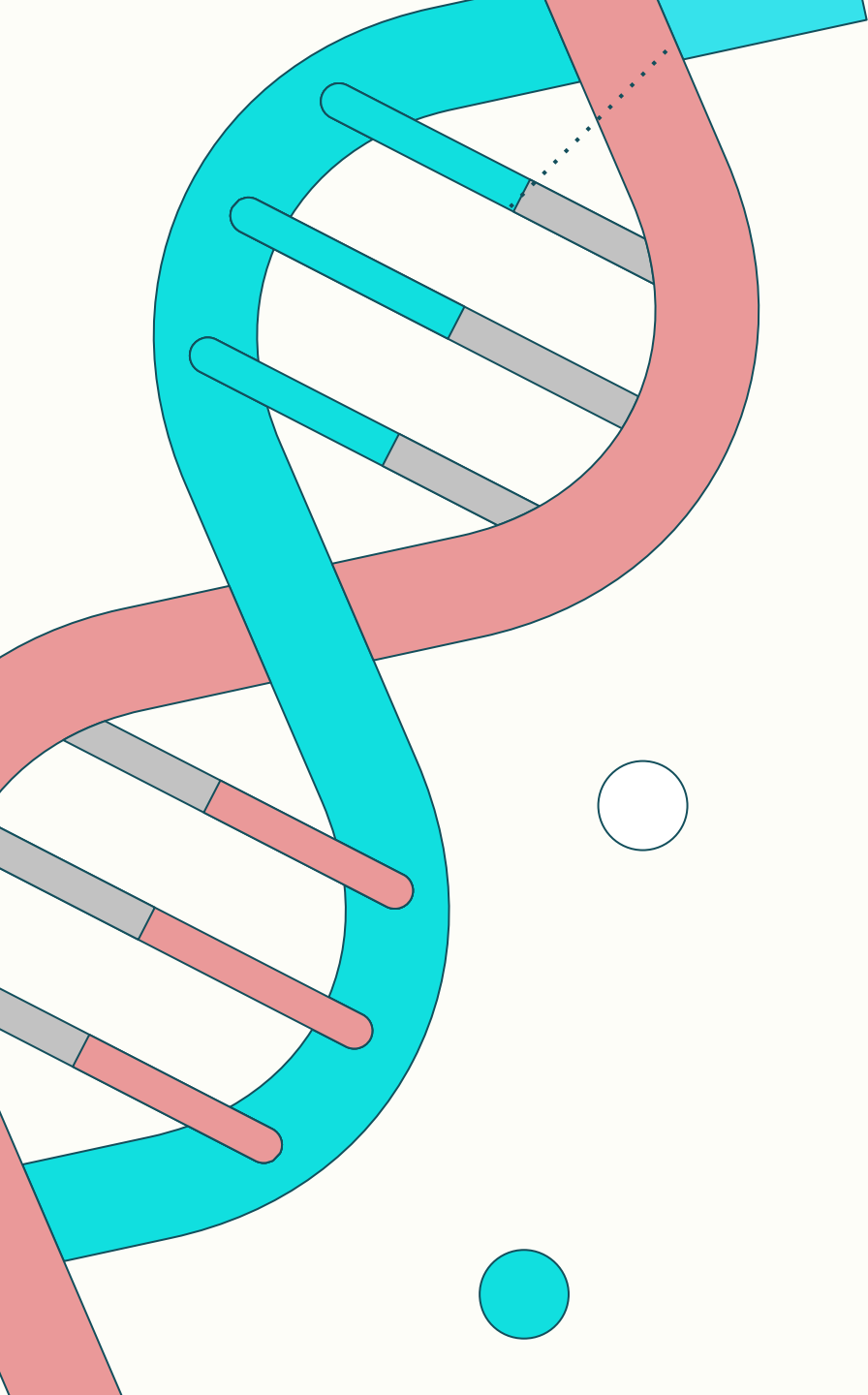




W Polsce jako dowód w sprawie karnej uznaje się profil DNA, jeśli zgodne jest przynajmniej 16 autosomalnych markerów + marker płci.

Wtedy prawdopodobieństwo, że dwa pełne profile będą przypadkowo zgodne wynosi 10^{-16} .

Obecnie standardowo w Unii Europejskiej analizowane jest 20 markerów.



Dziękuję za uwagę

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